

TRACK 2 · VIRTUAL POWER PLANT B2B

FlexGrid

Smart Enterprise Energy Orchestrator

We turn a company from a passive consumer into a **supplier of flexibility** for the grid and get it paid for that.

Team AI A/B Users – Federico · Alessandro · Davide · Manuel

316→186 kW
GRID DRAW

-130 kW
DELIVERED

80/20
REVENUE SPLIT

01 PROBLEM

The grid needs flexibility. Companies have it but they don't sell it.

THE MARKET

Terna pays for flexibility

On the Dispatching Services Market it rewards whoever reduces or raises their draw in real time.

THE ASSET

Companies are prosumers

Solar PV, EV chargers, HVAC, adjustable loads: the flexibility is already installed on site.

THE GAP

Orchestration is missing

That flexibility sits unused today: nobody aggregates it and brings it to market.

TECHNICAL CHALLENGE

Coordinate many distributed sites (edge) from a central brain (cloud), securely and at scale.

An intelligent aggregator

Modulates flexible loads on Terna's command. Three asset categories, one clear rule for each.
each.

01 · FLEXIBLE

EV smart-charging HVAC

Modulated ↓

02 · CRITICAL

Production Servers

Untouchable ✕

03 · GENERATION

Solar PV

Always at max ↑

ECONOMIC MODEL · REVENUE SPLIT

80% · Company

20% · Platform

How it works

Two FastAPI microservices on the edge–cloud continuum.

Terna signal POST /terna-event

CLOUD

Aggregator / Dispatch Engine · FastAPI

- Computes the dispatch split
- Accounts for the 80/20 economics
- Writes telemetry + € to InfluxDB

InfluxDB → Grafana

MQTT

← telemetry

commands →

Mosquitto

EDGE

Edge Controller + Simulator · FastAPI

- Governs PV / EV / HVAC / critical critical
- Executes commands, publishes telemetry
- 1 replica = 1 new site site

Docker → Kubernetes

FastAPI

MQTT / Mosquitto

InfluxDB + Grafana

Kubernetes · 2 namespaces

The role of OpenSlice

Orchestrates services on the continuum and makes them scalable like a catalog.

Docker
packages



Kubernetes
runs



OpenSlice
orchestrates & orders

REAL CLUSTER · CRIDGE

OpenSlice is wired to our Kubernetes: it sees and manages it as a resource.

Resource Inventory: `kind-cei · available`

A NEW SITE = AN ORDER

A **service order** on the catalog → OpenSlice deploys the edge service.

catalog → order → `deploy`

SINGLE PORTFOLIO

The flexibility of **N sites** sums into one product product sellable to Terna.

1 → N, `with one click`

A Terna event

	BEFORE	AFTER
Grid draw	316 kW	186 kW
EV chargers	176 kW	56 kW
HVAC	100 kW	90 kW
Critical loads	200 kW	200 kW ✓

–130 kW

reduction request

kW

FLEXIBILITY DELIVERED

32.5 kWh

EVENT REVENUE

€13

€10.40 company · €2.60 platform

Works today, scales tomorrow

IMPACT	More renewables integrated into the grid, companies paid for paid for assets they already own, decarbonization.	BUSINESS	Recurring revenue hundreds of activations per year plus the availability fee paid by Terna.
SCALABILITY	With OpenSlice, 1 site becomes N with one click on the catalog. catalog.	OPEN-SOURCE	100% open stack: FastAPI · Mosquitto · InfluxDB · Grafana · · Kubernetes · OpenSlice ETSI.

TECHNICAL AMBITION

Full OpenSlice running on the continuum, connected to a real Kubernetes cluster.

Thank you **for the attention.**